

Keywords: mechanistic interpretability • causal circuits • post-training • LLM fine-tuning

Circuit Condensation: Post-Training that Concentrates a Behavior’s Causal Circuit

SFT Universally Condenses Causal Circuits While RL Expands Multi-Step Reasoning Circuits

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arXiv:2608.27254 • Preprint, August 2026

Does post-training reorganise the internal circuits of a language model in interpretability-friendly ways? Circuit Condensation answers this question empirically across eight behaviours and three post-training regimes (SFT, GRPO, and base). Using ACDC and edge attribution patching, the paper shows that SFT consistently shrinks and focuses causal circuits (+20% faithfulness, −31% component count), while RL post-training condenses simple circuits but expands complex multi-step ones — a divergence with direct implications for scalable interpretability.

AT A GLANCE

Problem: It is unknown whether post-training concentrates or disperses the causal circuits responsible for specific behaviors, with direct consequences for scalable mechanistic interpretability.

Why it matters: If post-training concentrates circuits, automated circuit analysis becomes easier; if it disperses them, interpretability may degrade even as capability improves.

Method: ACDC and Edge Attribution Patching applied before and after SFT, GRPO, and combined post-training across 8 representative LLM behaviors.

Result: SFT: −31% circuit size, +0.15 edge faithfulness. GRPO: condenses factual circuits but expands multi-step reasoning circuits by +47%.

Evidence: Effect is robust across learning rates and dataset sizes for SFT; behavior-type (simple vs. multi-step) determines RL’s direction of effect.

The Big Picture

Mechanistic interpretability aims to identify the sparse subgraph of a neural network’s computation that causally produces a specific behaviour — its *circuit*. Progress in circuit discovery has revealed that transformers implement interpretable algorithms in relatively compact circuits: the indirect object identification circuit in GPT-2 involves ∼26 attention heads; the greater-than circuit involves ∼10.

But circuits are always identified in *base* models. Post-training dramatically changes model behaviour. A natural question follows: does post-training make circuits more concentrated (easier to identify and edit) or more dispersed (harder)? This question is not merely academic — interpretability tooling built for base models may perform very differently on fine-tuned models used in deployment. Circuit Condensation provides the first systematic answer.

Why Existing Approaches Fall Short

Static analysis: Prior mechanistic interpretability work analyses circuits in fixed (base or RLHF’d) models but does not compare circuit structure across training stages. There is no prior data on whether post-training increases or decreases circuit sparsity.

Limited behaviour coverage: Some work examines single behaviours (e.g., factual recall) before and after fine-tuning, but the generality of any observed effect is unclear. Circuit Condensation covers eight behaviours of varying complexity to identify patterns.

Confounded by capability: If a model becomes more accurate at a behaviour after fine-tuning, circuit quality may change even if circuit structure is unchanged. The paper controls for capability by measuring circuit faithfulness at matched accuracy levels.

Core Mechanism

Circuit discovery methods:

ACDC (Automated Circuit Discovery) iteratively ablates edges in the computation graph — replacing activations with corrupted versions from a distribution matched to the task — and retains an edge only if its ablation decreases performance by more than threshold ϵ . The result is a sparse *minimal circuit* sufficient to perform the behaviour.

Edge Attribution Patching (EAP) scores each edge by a first-order approximation:

$$\text{EAP}(e) = \frac{\partial \mathcal{L}}{\partial h_e} \cdot (h_e^{\text{clean}} - h_e^{\text{corrupt}})$$

where h_e is the activation at edge e , and $\mathcal{L}$ is the task loss. EAP enables fast ranking of all edges without exhaustive ablation.

The paper uses ACDC to identify circuits (trusted, slower) and EAP to validate patterns (fast, scalable to larger models).

Circuit metrics:

- **Circuit size:** number of attention heads and MLP layers with non-zero ACDC edges
- **Edge faithfulness:** fraction of task performance recovered when running only the circuit (with other

components ablated)

- **Modularity:** intersection-over-union of circuit components across behaviors (lower = more behavior-specific)

Technical Formulation

Let $C_b(\theta)$ be the ACDC circuit for behavior b under parameters θ . The paper tracks three metrics across θ :

$$\text{Size}(C_b) = |V(C_b)| + |E(C_b)|$$

$$\text{Faith}(C_b) = \frac{\text{Perf}_{C_b}}{\text{Perf}_{\theta}}$$

$$\text{Mod}(b_1, b_2) = 1 - \frac{|C_{b_1} \cap C_{b_2}|}{|C_{b_1} \cup C_{b_2}|}$$

where Perf_C is task performance with only circuit C active. Comparisons are made between θ_{base} , θ_{SFT} , and θ_{RL} .

Learning or Inference Procedure

For each of 8 behaviors, the paper trains three model versions:

1. Base model (no post-training)
2. SFT on 500 behavior-specific demonstrations
3. GRPO on the same 500 problems with scalar correctness rewards

Circuit discovery is run at each checkpoint using ACDC (threshold $\epsilon = 0.01$) on 100 held-out task instances. EAP is run on 500 instances to validate ACDC findings.

What the Guarantee Says

No formal convergence guarantees are given. The empirical claim is that the observed patterns (SFT condenses; RL behavior depends on task complexity) are robust across model sizes (1B–7B parameters tested), learning rates, and dataset sizes.

Experimental Findings

- **SFT (all behaviors):** Average circuit size decreases by -31% ; edge faithfulness increases from 0.74 to 0.89
- **GRPO (simple behaviors: factual recall, IOI):** Circuit size decreases -28% ; faithfulness $+0.11$ — similar to SFT
- **GRPO (multi-step behaviors: arithmetic, logical):** Circuit size *increases* $+47\%$; faithfulness *decreases* -0.06 — opposite direction from SFT
- **Modularity:** SFT increases modularity (circuits become more behavior-specific); GRPO has mixed effects

- **Combined SFT+GRPO:** SFT condensation is partially preserved after subsequent GRPO for simple behaviors, but overridden for multi-step behaviors

Ablations and Interpretation

Learning rate: SFT condensation is robust across learning rates from 10^{-5} to 10^{-3} ; the magnitude of size reduction varies slightly but direction is consistent.

Dataset size: Condensation occurs even with as few as 50 demonstrations; more demonstrations yield larger faithfulness gains.

Why does RL expand multi-step circuits? The paper hypothesises that RL via GRPO improves multi-step performance by recruiting additional attention heads to handle credit propagation over long chains. This is consistent with findings from other work showing that RL fine-tuning on reasoning tasks induces new attention head specialisations.

Implications for circuit-based editing: For SFT-trained models, targeted activation patching and component-level editing become more reliable because circuits are more concentrated and faithful. For RL-trained models used on complex reasoning tasks, the expanded circuits make editing harder — a previously unexplored cost of RL post-training.

Reference

Sai Adith Senthil Kumar. **Circuit Condensation: Post-Training that Concentrates a Behavior’s Causal Circuit.** arXiv:2608.27254, August 2026. <https://arxiv.org/abs/2608.27254>